

AI Diabetes Risk Predictor

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1. Project Overview

The goal of this project is to demonstrate how AI can be used to support early risk assessment of diabetes using structured medical data.

The application allows users to enter patient information such as glucose level, blood pressure, insulin level, BMI, age and other health-related features. Based on these values, the trained machine learning model returns a diabetes risk prediction together with a probability score and a simple explanation of risk factors.

2. Dataset

The project uses the [Diabetes Dataset](#) from Kaggle. The dataset contains tabular medical data with the following features:

- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age
- Outcome

The target column is Outcome, where 0 means no diabetes, and 1 means diabetes.

Before training the model, unrealistic zero values in selected columns such as Glucose, BloodPressure, SkinThickness, insulin and BMI were replaced with median values. This preprocessing step helped in improving the quality of the data and made the model more reliable.

3. Technologies used

This project was built using the following technologies:

- Python- as main programming language
- pandas- data loading and preprocessing
- scikit-learn- machine learning model training and evaluation
- Streamlit- interactive web application
- matplotlib- data visualization
- pickle- saving and loading the trained model and scaler

4. Machine Learning Model

The application uses a Random Forest Classifier to predict diabetes risk. Random Forest was chosen because it works well with tabular data and allows feature importance analysis, which helps make the model more explainable.

To improve the model performance, GridSearchCV was used to test different model parameters and to select the best combination. The model was optimized mainly for recall, since in healthcare, false negatives can be ore serious than false positives. Missing a patient who may be at risk of diabetes is more problematic than incorrectly flagging the one who is a low-risk patient.

A custom prediction threshold of 0.40 was used to increase sensitivity for detecting diabetes risk.

5. Application features

Prediction

This section allows the user to enter health data and receive a diabetes risk prediction. The result is displayed as:

- Low risk
- Moderate risk
- High risk

The app also shows the risk probability and a simple explanation of possible risk factors such as high BMI or glucose levels.

Patient Information

Pregnancies

1

- +

Insulin



Glucose level



BMI



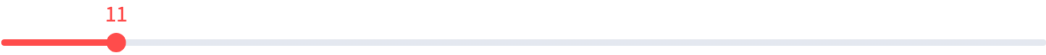
Blood Pressure



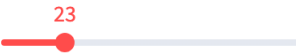
Diabetes Pedigree Function



Skin Thickness



Age



Predict Diabetes Risk

Moderate risk of diabetes

Risk probability

38.96%

Risk level

Moderate risk

Dataset Overview

This section presents basic information about the dataset, including the number of rows, columns, diabetes cases, sample data, outcome distribution and descriptive statistics.

Dataset Overview ↗

The model is trained on the Pima Indians Diabetes Dataset, which contains tabular medical features used for binary diabetes classification.

Rows

768

Columns

9

Sample Data

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	Diabetes
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

Model Insights

This section explains how the model works and presents evaluation metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

The confusion matrix helps with showing how many cases were correctly or incorrectly classified. Special attention is given to false negatives, because in a healthcare context it is important to reduce the number of patients with diabetes risk who are incorrectly predicted as low risk.

Model Insights

The final model uses a Random Forest Classifier. In healthcare-related classification, recall for the positive class is especially important b
producing a false alarm.

Model Performance Metrics

The metrics below are calculated using the current trained model and the diabetes dataset loaded in the app. A custom threshold of 0.40

Accuracy

0.75

Precision

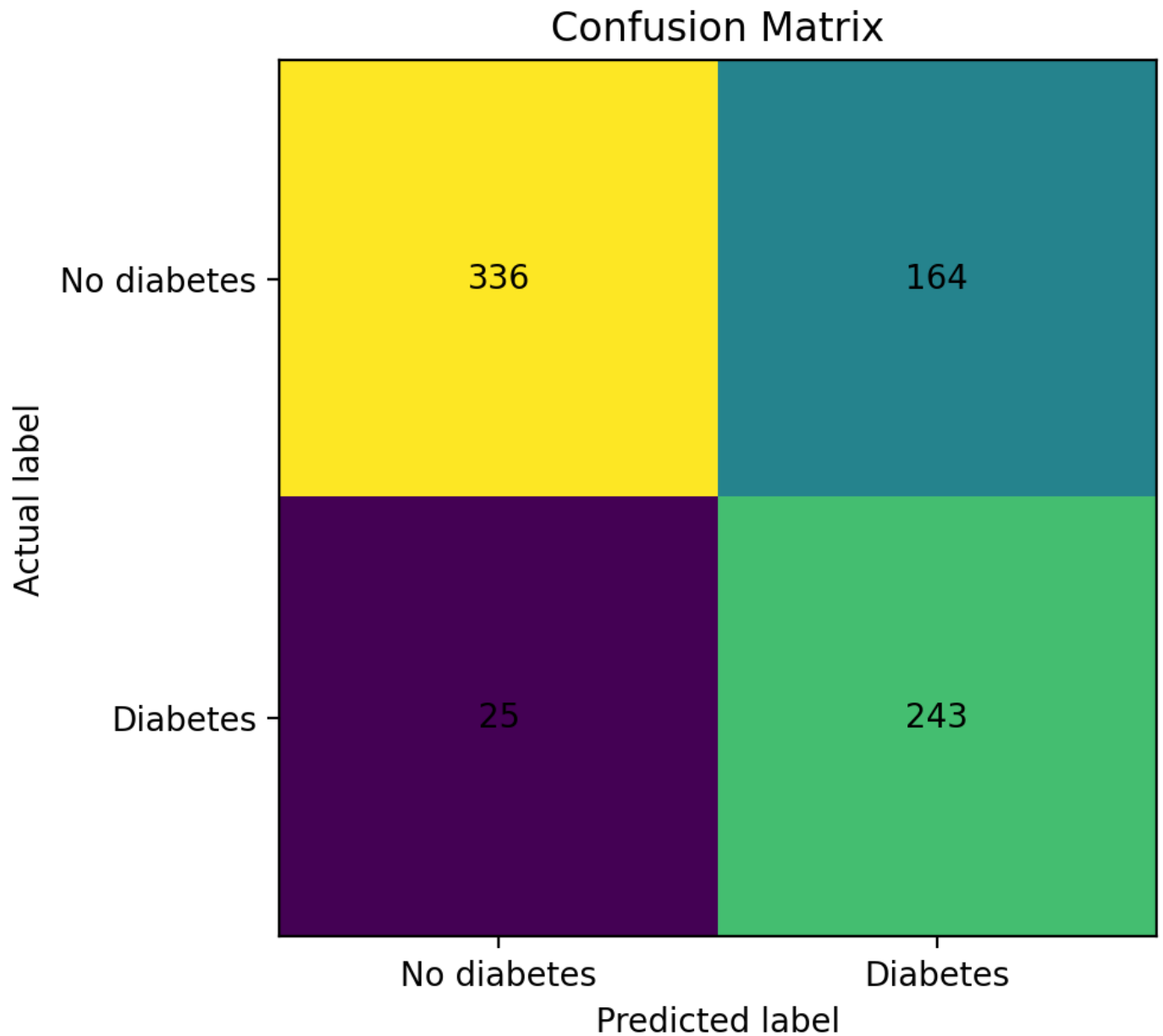
0.60

Recall

0.91

Confusion Matrix

	Predicted: No diabetes	Predicted: Diabetes
Actual: No diabetes	336	
Actual: Diabetes	25	



6. Conclusion

AI diabetes Risk Predictor demonstrates how machine learning can be applied to healthcare-related data to support diabetes risk prediction. The project combines data preprocessing, model training, evaluation and an interactive web interface.

It shows my practical skills in Python, machine learning and data analysis. At the same time, it highlights responsible AI use in healthcare and the need to treat the model as supportive information rather than medical decisions.

